# MIRZA AGHA MALIK BAIG

**Computer Engineer**

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## Summary

M.S. Computer Engineering student focused on Design Verification with hands-on experience in SystemVerilog, UVM, constrained-random testing, functional coverage, and AMBA APB/AXI4 verification.

## Technical Skills

- **Tools and Software:** Synopsys VCS, DVE, QuestaSim/ModelSim, Xilinx Vivado, GTKWave, EDA Playground, Git, GitHub.
- **Programming and HDLs:** SystemVerilog, Verilog, Python, C.
- **Verification Methodologies:** UVM, Constrained Random Verification (CRV), Directed Testing, Coverage-Driven Verification.
- **Key Concepts:** RTL Design, RTL Simulation, Object-Oriented Programming, SystemVerilog Assertions, Code Coverage, Functional Coverage, Constraints, Randomization, Scoreboards, Testbench Development, Waveform Debugging.
- **Protocols and Architecture:** AMBA APB, AXI4, SPI, I2C, UART, FIFO, Pipelining, Data/Control Hazards, Memory Hierarchy, ASIC Design Flow.

## Experience

**Aisemi Pvt. Ltd., Verification Intern** *Jan 2024 – Apr 2024*

- Completed a 4-month internship focused on SystemVerilog, RTL verification fundamentals, and semiconductor concepts.
- Developed and simulated SystemVerilog RTL and testbenches for combinational and sequential digital logic.
- Used Synopsys tools to run simulations, inspect waveforms, and debug functional mismatches.

**Electronics Corporation of India Limited (ECIL), Industrial Trainee** *Nov 2019 – Mar 2020*

- Completed industrial training at ECIL, a Government of India enterprise, with exposure to electronic systems, digital circuits, and laboratory practices.
- Performed supervised circuit testing, signal observation, measurement, and basic hardware troubleshooting.

## Academic Projects

**Design and UVM Verification of APB Memory**

- Designed and verified an AMBA APB memory slave using UVM with constrained-random stimulus, scoreboard, assertions, and functional coverage.
- Developed self-checking tests for reset, read/write transfers, boundary values, randomized address/data, repeated accesses, and write-followed-by-read data integrity.
- Debugged protocol and data mismatches using scoreboard failures and VCS waveforms, correcting RTL/testbench issues before coverage closure.
- Achieved 100% functional coverage, 100% line and branch coverage, 70.43% toggle coverage, and 60% condition coverage.

**Design and UVM Verification of AXI4 Memory**

- Developed a UVM verification environment for a memory-mapped AXI4 slave covering AW, W, B, AR, and R channels.
- Verified VALID/READY handshakes, independent AW/W timing, backpressure, WSTRB, responses, reset behavior, and read-after-write data integrity.
- Implemented scoreboard-based memory checking and functional coverage for transactions, handshakes, address/data values, responses, and protocol scenarios.

**SystemVerilog FIFO Verification Environment**

- Developed a class-based verification environment for a parameterized synchronous FIFO using transaction, generator, driver, monitor, and scoreboard components.
- Verified read/write, simultaneous operations, full/empty, overflow/underflow, and boundary conditions using directed/randomized tests, coverage, assertions, and waveform analysis.

## Education

**California State University, Northridge (CSUN)** *Aug 2024 – Dec 2026*

*M.S. in Computer Engineering*

*Relevant Coursework:* Verilog & SystemVerilog, ASIC Development, System-on-Chip Design, Computer Systems Architecture, Computer Arithmetic Design

**Osmania University** *Jul 2021 – Jun 2023*

*B.E. in Electronics & Communication Engineering*

*Relevant Coursework:* Digital Logic Design, VLSI Design, Microprocessors & Microcontrollers, Embedded Systems